

HOMEWORK ASSIGNMENT 1

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

Instructions:

- Kindly recall that homework assignments for this course are **not turned in**, and **not graded**.
 - However, problems from homework sets will form the basis of problems that appear on the weekly quizzes; as such, it is in your best interest to complete the homework sets on your own
 - Solutions to this assignment will be released at 8:00am on Thursday, August 27 in time for the corresponding quiz on Monday August 31
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1. **Variable Classification.** Classify each of the following variables as either categorical or quantitative.

- (a) The amount of time (in minutes) it takes a student to get to Cal Poly from their place of residence.
☐ Categorical ☐ Quantitative
- (b) The mode of transportation (car, bike, bus, etc.) a student uses to get to Cal Poly from their place of residence.
☐ Categorical ☐ Quantitative
- (c) The proportion of the day a Cal Poly student spends on their phone.
☐ Categorical ☐ Quantitative
- (d) The number of classes a Cal Poly student is enrolled in this semester.
☐ Categorical ☐ Quantitative
- (e) A Cal Poly student's favorite coffee shop in town
☐ Categorical ☐ Quantitative

2. **A Picture is Worth a Thousand Words.** For each of the following scenarios, identify the plot or plots that are most appropriate. If there are multiple potentially appropriate plots, try to list them all.

- (a) Yuri has collected data on the speeds (in miles per hours) of several cars traveling along Highway 101, and wants to make a plot that displays the distribution of speeds.
- (b) Qing has surveyed 100 people and asked them what brand of bottled water they most prefer. She now wants to make a plot that displays the distribution of bottled water brand preferences.
- (c) Miguel wants to determine whether or not there is a relationship between the amount of sleep students get the night before an exam and their score on the exam. Specifically, he would like to use an appropriate plot to make this determination.
- (d) Puja has surveyed the students in her class, and asked them which month they were born in. She would like to visualize the distribution of birth months, to determine which months are most popular.

3. **Living Life to the Fullest.** The *World Bank* is a collection of several agencies, dedicated to studying the global effects of poverty. As a part of their mission, they provide open source data on various attributes of countries around the globe. For the purposes of this problem, we'll focus on one particular attribute: life expectancy. Each data point represents the total life expectancy (in years) of a country, as collected in 2024. Shown below is a histogram of the data:

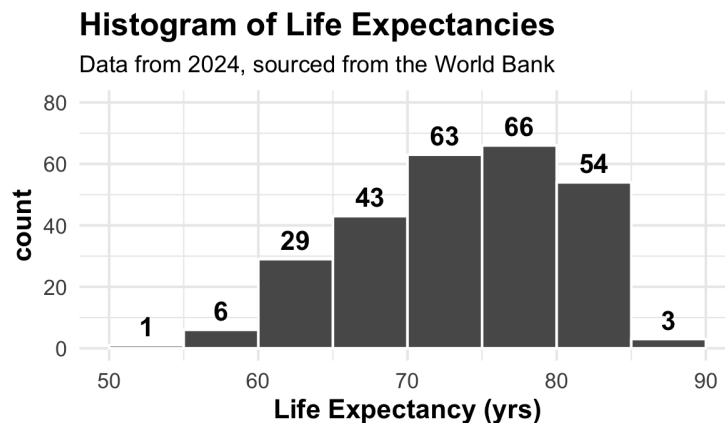


Figure 1: Histogram of Life Expectancy (in years) of 265 countries. Data has been sourced from the World Bank Open Data Repository¹

- (a) What are the observational units in this dataset?
- (b) Describe the distribution of life expectancies. Remember that there are four main characteristics you should describe: shape, center, variability, and outliers. Further recall that your description of “shape” should include mention of the number of modes, symmetry, and skew.
- (c) Do you think the median life expectancy is higher, lower, or about the same as the mean life expectancy? Why?
- (d) What proportion of countries in the dataset have life expectancies between 70 and 80 years?
4. **Follow the Numbers!** A list of twelve numbers has sample mean equal to 4 and median equal to 4.6.
- (a) **True or False:** the number 4 must have been one of the original twelve numbers.
☐ True ☐ False
- (b) **True or False:** the number 4.6 must have been one of the original twelve numbers.
☐ True ☐ False
- (c) What was the sum of the twelve numbers? How do you know?
- (d) A new (thirteenth) number is incorporated to the list, and it is equal to 3.25. What is the mean of all thirteen numbers?

¹<https://data.worldbank.org/>